

Student Name:

Assessment Number:

Teacher's Name:



PLC PRESBYTERIAN
LADIES' COLLEGE
SYDNEY

1888

2019
TRIAL
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics

General Instructions

- Reading time - 5 minutes
- Working time - 3 hours
- Write using **black pen**
- Board-approved calculators may be used
- A multiple choice answer sheet is attached to the examination paper
- All necessary working should be shown in every question
- A reference sheet is provided

Total Marks - 100

Section I:

10 marks

- Attempt questions 1-10, using the multiple choice answer sheet. Allow about 15 minutes for this section

Section II:

90 marks

- Attempt questions 11-16, using the Answer Booklets provided.
- Allow about 2 hours 45 minutes for this section.

Multiple Choice	11	12	13	14	15	16	Total
/10	/15	/15	/15	/15	/15	/15	%

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Section I

10 marks

Attempt Questions 1 – 10.

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 – 10.

1. What is the value of $4\pi \sqrt{\frac{a}{g}}$, correct to two significant figures, where $a = 4.1$ and $g = 9.8$?

- (A) 2.6
- (B) 5.3
- (C) 8.1
- (D) 19



- What are the possible values of x for the inequality $1 - 3x > 10$?

- (A) $x > -3$
- (B) $x < -\frac{11}{3}$
- (C) $x > -\frac{11}{3}$
- (D) $x < -3$

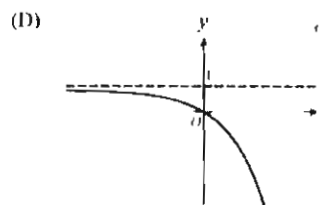
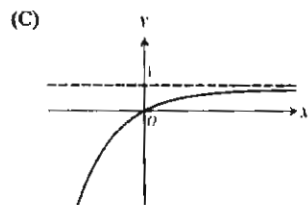
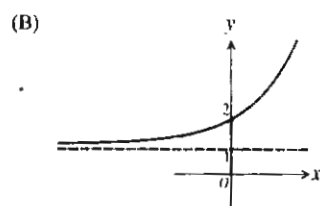
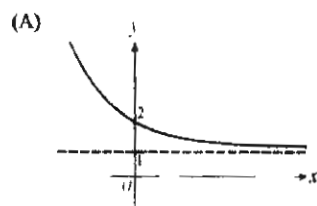
3. What is the derivative of $\ln(\sin x)$?

- (A) $\tan x$
- (B) $\operatorname{cosec} x$
- (C) $\cot x$
- (D) $\ln(\cos x)$

4. For the angle θ , $\tan \theta = \frac{9}{40}$ and $\sin \theta = -\frac{9}{41}$. What is the value of $\cos \theta$?

- (A) $-\frac{9}{40}$
 (B) $\frac{40}{41}$
 (C) $-\frac{40}{41}$
 (D) $\frac{41}{40}$

5. Which is the correct graph for $y = 2^{-x} + 1$?



6. The point $P(x, y)$ moves such that it remains equidistant from two fixed points. Which of the following equations might describe the locus of P ?

- (A) $3x + 2y - 5 = 0$
 (B) $(x - 1)^2 + (y + 3)^2 = 9$
 (C) $y = \frac{3}{x}$
 (D) $x^2 = 12y$

7. Given $\log_x 3 = 0.6$ and $\log_x 2 = 0.4$, find $\log_x 24$.

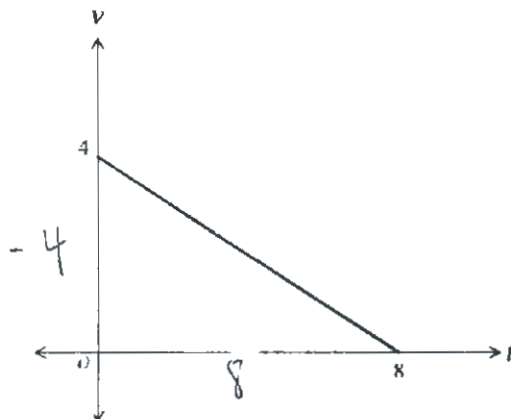
- (A) 0.0384
 (B) 0.24
 (C) 0.72
 (D) 1.8

8. What is the domain and range of the function $f(x) = |x - 5| + 2$?

- (A) Domain: $x \geq 0$, Range: $y \geq 0$
 (B) Domain: $x \geq 5$, Range: $y \geq 2$
 (C) Domain: All real x , Range: $y \geq 2$
 (D) Domain: All real x , Range: $y \geq -3$

9

The graph below shows the velocity of a particle as it moves along the horizontal axis. The velocity is measured in metres per second and time is measured in seconds.



When $t = 0$ the particle's displacement, x , is -2 metres.
What is the maximum value of the particle's displacement?

- (A) 4 m
- (B) 12 m
- (C) 14 m
- (D) 32 m

10. Consider a curve with the following properties:

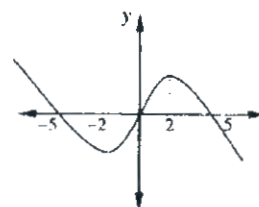
$g(x)$ is odd

$$g(5) = 0 \text{ and } g'(2) = 0$$

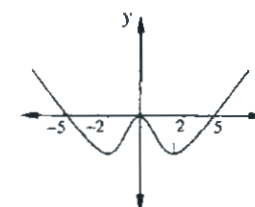
$$g'(x) > 0 \text{ for } x > 2$$

Which of the following could be the graph of $y = g(x)$?

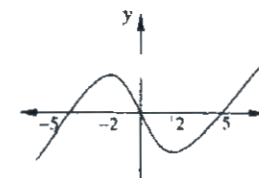
(A)



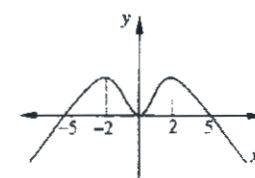
(B)



(C)



(D)



End of Section I

Section II

90 marks

Attempt Questions 11 – 16.

Allow about 2 hours and 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

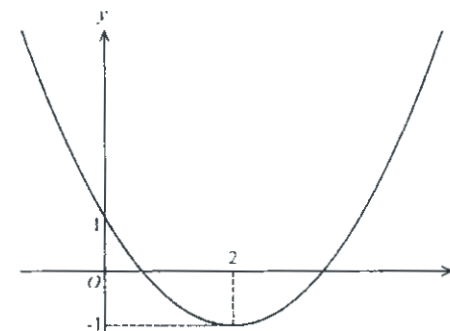
Question 11 (15 marks) Use a new writing booklet.

- (a) Rationalise the denominator of $\frac{2}{\sqrt{3}-2}$. 2
- (b) Show that the result of differentiating $\frac{2x^3}{x-2}$, with respect to x , is $\frac{4x^2(x-3)}{(x-2)^2}$. 2
- (c) Find $\int \frac{x^2}{3x^3-1} dx$. 2
- (d) The gradient of interval AB is 5, where A and B are the points $(x, 2)$ and $(-4, y)$ respectively. Find one possible value for x and one possible value for y . 2

Question 11 continues on the next page

Question 11 continued

- (e) The diagram shows the graph of a parabola passing through the point $(0, 1)$ and has its vertex at $(2, -1)$.



- (i) Find the equation of the parabola. 2
- (ii) Write down the coordinates of the focus and the equation of the directrix. 2

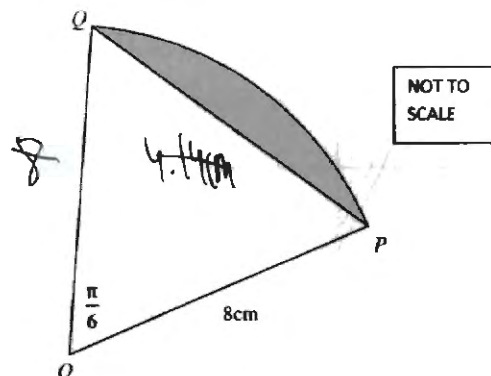
Question 11 continues on the next page

Question 11 continued

- (f) A sector OPQ of a circle, is shown in the diagram.

The circle has its centre at O and a radius of 8 cm.

$$\angle QOP = \frac{\pi}{6}$$



- (i) Find the length of the interval PQ , correct to 2 decimal places.

2

- (ii) Calculate the perimeter of the shaded segment, correct to 1 decimal place.

1

End of Question 11

Question 12 (15 marks) Use a new writing booklet.

- (a) The quadratic function $2x^2 - x - 15$ has roots α and β . Find the value of:

3

(i) $\alpha + \beta$

(ii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$

- (b) A straight line with equation $y = mx + b$ is a tangent to the curve with equation $y = x^2 + 6x + 7$.

(i) By using the discriminant, show that $m^2 - 12m + 4b + 8 = 0$.

2

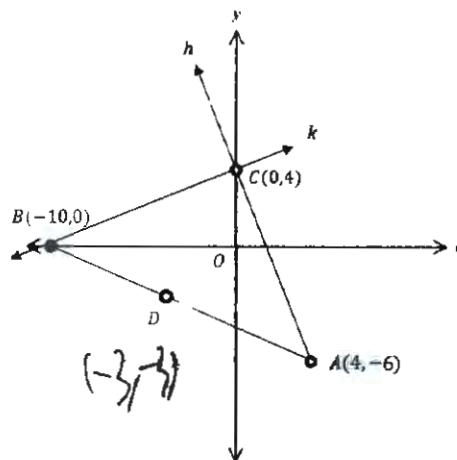
(ii) Given $4b + m^2 = 0$, find the y -intercept of the tangent to the curve.

1

Question 12 continues on the next page

Question 12 continued

- (c) In the diagram, the lines h and k are drawn. The coordinates of A , B and C are $(4, -6)$, $(-10, 0)$ and $(0, 4)$ respectively. D is the midpoint of AB .



- | | |
|---|---|
| (i) Show that D has coordinates $(-3, -3)$. | 1 |
| (ii) Calculate the length of DC . Leave your answer in exact form. | 1 |
| (iii) Show that the equation of the line h is given by $5x + 2y - 8 = 0$. | 2 |
| (iv) Show that the line h is perpendicular to the line k . | 2 |
| (v) AB is the diameter of a circle which passes through the points A , B and C . Find the equation of the circle. | 2 |
| (vi) Find the area of the circle which passes through A , B and C . Leave your answer in terms of π . | 1 |

End of Question 12

Question 13 (15 marks) Use a new writing booklet.

- | | |
|--|---|
| (a) Solve: $\log_e(x^2 + 9) - \log_e(5x) = \log_e 2$. | 2 |
| (b) Consider the curve $y = 2x^3 - 9x^2 + 12x - 5$. | |
| (i) Find the stationary points of the curve and determine their nature. | 3 |
| (ii) Given that the point $(1.5, -0.5)$ lies on the curve, show that it is a point of inflexion. | 1 |
| (iii) Sketch the curve, labelling its important features. | 2 |
| (c) Find $\lim_{x \rightarrow 2} \frac{4x - x^3}{2 - x}$. | 2 |

Question 13 continues on the next page

Question 13 continued

- (d) Australia's population was 19 million at the start of the year 2000. By the beginning of 2019, it had grown to 25 million.
- Assuming that the population of Australia is increasing exponentially, it can be represented by an equation in the form $P = Ae^{kt}$, where A and k are constants, and t is measured in years from the beginning of 2000.
- (i) Show that $P = Ae^{kt}$ satisfies the equation $\frac{dP}{dt} = kP$. 1
- (ii) What is the value of A ? 1
- (iii) Show that $k \approx 0.0144$. 2
- (iv) A newspaper article, published in 1980, forecast that the Australian population would reach 50 million by the year 2050. Using calculations to justify your response, state whether you agree or disagree with this article. 1

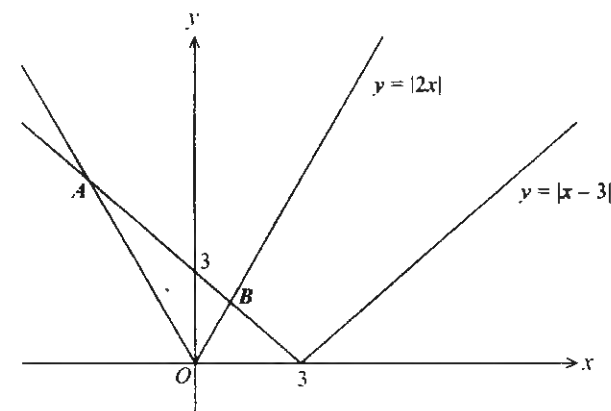
End of Question 13

Question 14 (15 marks) Use a new writing booklet.

- (a) Find the equation of the tangent to the curve $f(x) = e^{1-2x}$ at the point $\left(\frac{1}{2}, 1\right)$. 2

- (b) The sides of a triangle are 7 cm, 5 cm and 4 cm. Find the size of the angle opposite the longest side in radians, correct to 2 decimal places. 1

- (c) The diagram shows the graphs of $y = |2x|$ and $y = |x - 3|$ which intersect at points A and B .



- (i) Find the x -coordinate of points A and B . 2
- (ii) Hence or otherwise, solve $|x - 3| \leq |2x|$. 1

Question 14 continues on the next page

Question 14 continued

- (d) The flow rate of water first into and then out of a fountain is R where $R = 3\pi t(8 - t)$ litres/second.

(i) When does the water stop flowing into the fountain?

1

(ii) The fountain was initially empty. At what time is the fountain empty again?

2

(iii) What is the exact rate at which the fountain was losing water at that time?

1

- (e) The displacement of a particle moving along the x -axis is given by

$$x = t + \ln(3t + 1)$$

where t is the time in seconds and x is measured in centimetres (cm).

(i) Show that the particle never comes to rest.

2

(ii) Find the distance travelled by the particle during the 3rd second. Give the answer correct to 1 decimal place.

1

(iii) Write down an expression for the acceleration of the particle

1

(iv) Is the particle slowing down or speeding up for $t > 0$? Give reasons to support your answer.

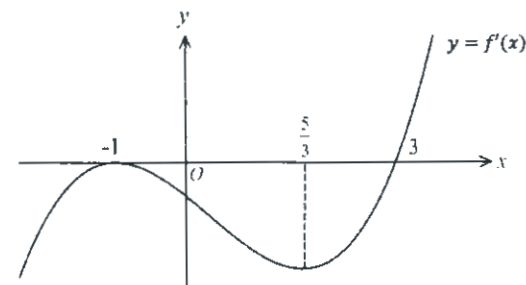
1

End of Question 14

Question 15 (15 marks) Use a new writing booklet.

- (a) The diagram shows the graph of $y = f'(x)$ for the curve $y = f(x)$.

The curve of $y = f(x)$ has two stationary points at $x = -1$ and $x = 3$.



Sketch a possible graph for the curve $y = f(x)$ given $f(0) = 0$.

2

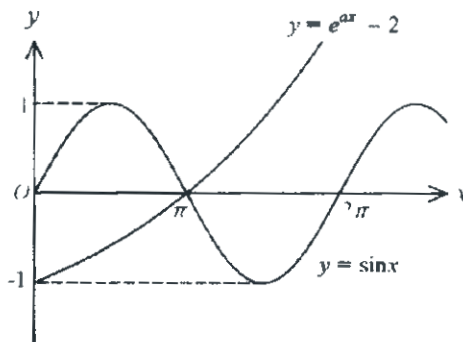
- (b) Solve the equation $2 \cos^2 x + \sin x = 2$, where $0 \leq x \leq 2\pi$.

3

Question 15 continues on the next page

Question 15 continued

- (c) The diagram shows the graphs of $y = \sin x$ and $y = e^{ax} - 2$ for $x \geq 0$, where $a > 0$. The graphs intersect at $(\pi, 0)$.



(i) Show that $a = \frac{\ln 2}{\pi}$.

2

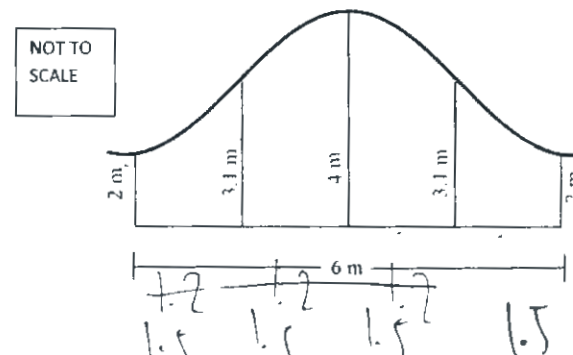
- (ii) Find the shaded area correct to 3 significant figures.

3

Question 15 continues on the next page

Question 15 continued

- (d) The diagram shows the front view of a farm shed. The area of the front of the shed has been split into 4 sub-intervals of equal width and their heights are shown.



- (i) Use two applications of Simpson's Rule to approximate the area of the front of the shed.

1

- (ii) Use four applications of the Trapezoidal rule to approximate the area of the front of the shed.

1

- (iii) The front of the shed has an area that is equivalent to the area between the curve $y = 3 + 3\cos x$, the x -axis and the lines $x = -3$ and $x = 3$. Calculate this area, correct to 2 decimal places.

2

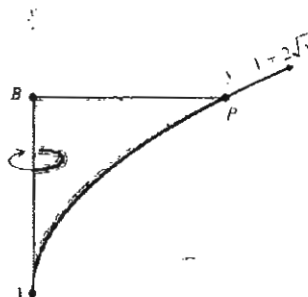
- (iv) Explain why both the Simpson's and Trapezoidal rule method give very similar approximations for the area.

1

End of Question 15

Question 16 (15 marks) Use a new writing booklet.

- (a) The shaded region in the diagram is bounded by the curve $y = 1 + 2\sqrt{x}$, the y -axis and the horizontal interval BP . The x -coordinate of point P is 4.



- (i) Find the coordinates of the point B . 1
- (ii) Calculate the exact volume of the solid of revolution formed when the shaded region is rotated about the y -axis. 3

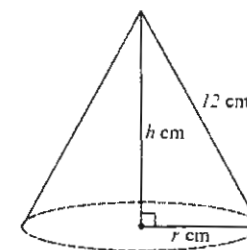
- (b) A curve $f(x)$ is known to have a gradient function $f'(x) = \sin^2 x \cos x$. 2
- It is also known that this curve passes through the point $(\frac{\pi}{2}, 1)$.
- Using the fact that $\frac{d}{dx}(\sin^3 x) = 3\sin^2 x \cos x$, find the equation for this curve.

Question 16 continues on the next page

Question 16 continued



The diagram shows a cone with height h cm, radius r cm and slant height of 12 cm.



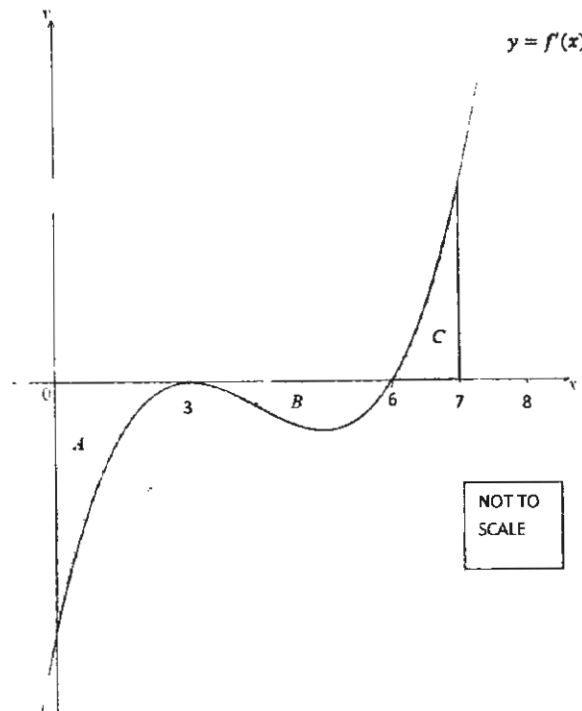
NOT TO SCALE

- (i) Show that the volume of the cone is given by $V = 48\pi h - \frac{1}{3}\pi h^3$. 1
- (Volume of a cone $V = \frac{1}{3}\pi r^2 h$)
- (ii) Find the maximum volume of the cone. Give the answer correct to 1 decimal place. 3

Question 16 continues on the next page

Question 16 continued

- (d) The following diagram shows the shaded regions A, B and C.



The regions are enclosed by the graph of $f'(x)$, the x -axis, the y -axis, and the line $x = 7$. The area of region A is 10, the area of region B is 3.65 and the area of region C is 3.65.

- (i) Given that $f(0) = 12$, find $f(7)$.

2

- (ii) Let $g(x) = (f(x))^2$. Given that $f'(7) = 14$, find the equation of the tangent to the graph of g at the point where $x = 7$.

3

End of Exam

Academic Year	12	Calendar Year	2019
Course	2 Unit	Name of task/exam	Trial

Section 1

1. $4\pi \sqrt{\frac{41}{9.8}} = 8.12809...$

$\therefore \textcircled{C} 8.1$

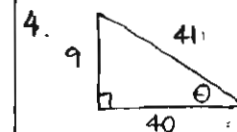
2. $1 - 3x > 10$

$-3x > 9$

$x < -3 \quad \textcircled{D}$

3. $\frac{1}{\sin x} \cdot \cos x = \frac{\cos x}{\sin x}$

$= \cot x \quad \textcircled{C}$

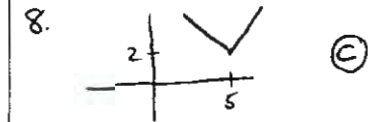


$\cos \theta = -\frac{40}{41} \quad \textcircled{C}$

5. $\textcircled{A}$

6. $\textcircled{A}$

7. $\log_x 24 = 3 \log_x 2 + \log_x 3$
 $= 3 \times 0.4 + 0.6$
 $= 1.8 \quad \textcircled{D}$



9. Max disp. when $v=0$
 i.e. $t=8$

gradient. $= -\frac{4}{8} = -\frac{1}{2}$

y-int $= 4$

$\therefore$ eqn of v is $v = -\frac{1}{2}t + 4$

$x = -\frac{t^2}{4} + 4t + c$

at $t=0$, $x=2$

$2 = c$

$\therefore x = -\frac{t^2}{4} + 4t + 2$

at $t=8$, $x = -\frac{64}{4} + 32 + 2$
 $= 14 \quad \textcircled{C}$

10. $\textcircled{C}$

Academic Year	12	Calendar Year	2019
Course	2 Unit	Name of task/exam	Trial

Section 2

Question 11

$$\begin{aligned} \text{a) } \frac{2}{\sqrt{3}-2} \times \frac{\sqrt{3}+2}{\sqrt{3}+2} &= \frac{2\sqrt{3}+4}{3-4} \\ &= \frac{2\sqrt{3}+4}{-1} \\ &= -2\sqrt{3}-4 \end{aligned}$$

$$\begin{aligned} \text{b) } u &= 2x^3 \quad v = x-2 \\ u' &= 6x^2 \quad v' = 1 \end{aligned}$$

$$\begin{aligned} \frac{d}{dx} &= \frac{6x^2(x-2) - 2x^3}{(x-2)^2} \\ &= \frac{6x^3 - 12x^2 - 2x^3}{(x-2)^2} \\ &= \frac{4x^3 - 12x^2}{(x-2)^2} \\ &= \frac{4x^2(x-3)}{(x-2)^2} \end{aligned}$$

$$\begin{aligned} \text{c) } f(x) &= 3x^3 - 1 \\ f'(x) &= 9x^2 \\ \therefore \frac{1}{9} \int \frac{9x^2}{3x^3 - 1} dx \\ &= \frac{1}{9} \ln(3x^3 - 1) + C \end{aligned}$$

$$\text{d) } \frac{y-2}{-4-x} = \frac{5}{1}$$

$$\begin{aligned} y-2 &= 5 & -4-x &= 1 \\ y &= 7 & -x &= 5 \\ & & x &= -5 \end{aligned}$$

$$\begin{aligned} \text{e) (i) } (x-h)^2 &= 4a(y-k) \\ \text{vertex } (2, -1) \end{aligned}$$

$$\begin{aligned} \therefore (x-2)^2 &= 4a(y-(-1)) \\ (x-2)^2 &= 4a(y+1) \end{aligned}$$

$$\text{Sub } x=0, y=1$$

$$\begin{aligned} (0-2)^2 &= 4a(1+1) \\ 4 &= 8a \end{aligned}$$

$$a = \frac{1}{2}$$

$$\begin{aligned} \therefore (x-2)^2 &= 2(y+1) \\ (x-2)^2 &= 2y+2 \end{aligned}$$

$$\begin{aligned} \text{(ii) focus } (2, -1 + \frac{1}{2}) \\ &= (2, -\frac{1}{2}) \end{aligned}$$

$$\begin{aligned} \text{directrix: } y &= -1 - \frac{1}{2} \\ y &= -\frac{3}{2} \end{aligned}$$

Academic Year	12	Calendar Year	2019
Course	2 Unit	Name of task/exam	Trial

11 f) (i)

$$\begin{aligned} PQ^2 &= 8^2 + 8^2 - 2 \times 8 \times 8 \cos \frac{\pi}{6} \\ &= 128 - 128 \times \frac{\sqrt{3}}{2} \\ &= 128 - 64\sqrt{3} \\ PQ &= 4.14 \text{ cm (to 2 dp)} \end{aligned}$$

$$\begin{aligned} \text{(ii) Arc length } PQ &= r\theta \\ &= 8 \times \frac{\pi}{6} \\ &= \frac{4\pi}{3} \end{aligned}$$

$$\begin{aligned} \text{Perimeter} &= \frac{4\pi}{3} + 4.14 \\ &= 8.32879... \\ &\approx 8.3 \text{ (to 1 dp)} \end{aligned}$$

Question 12

$$\begin{aligned} \text{a) (i) } \alpha + \beta &= -\frac{b}{a} \\ &= -\frac{(-1)}{2} \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{(ii) } \frac{\alpha}{\beta} + \frac{\beta}{\alpha} &= \frac{\alpha^2 + \beta^2}{\alpha\beta} \\ &= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} \\ \alpha\beta &= \frac{c}{a} = \frac{(-15)}{2} \\ &= \frac{(\frac{1}{2})^2 - 2 \times (-\frac{15}{2})}{-\frac{15}{2}} \\ &= \frac{\frac{1}{4} + 15}{-\frac{15}{2}} \\ &= -\frac{61}{30} \text{ or } -2\frac{1}{30} \\ &\text{or } -2.0\bar{3} \end{aligned}$$

$$\begin{aligned} \text{b) (i) Point of intersection} \\ x^2 + 6x + 7 &= mx + b \\ x^2 + 6x - mx + 7 - b &= 0 \\ \Delta = b^2 - 4ac &= 0 \text{ (meet at 1 point)} \\ (6-m)^2 - 4(7-b) &= 0 \\ 36 - 12m + m^2 - 28 + 4b &= 0 \\ m^2 - 12m + 4b + 8 &= 0 \end{aligned}$$

$$\begin{aligned} \text{(ii) } 4b + m^2 &= 0 \\ 4b &= -m^2 \\ b &= -\frac{m^2}{4} \\ \therefore m^2 - 12m + 4 \times (-\frac{m^2}{4}) + 8 &= 0 \\ m^2 - 12m - m^2 + 8 &= 0 \\ -12m + 8 &= 0 \\ m &= \frac{2}{3} \\ \therefore b &= -(\frac{2}{3})^2 \times \frac{1}{4} \\ &= -\frac{1}{9} \end{aligned}$$

Academic Year	12	Calendar Year	2019
Course	2Unit	Name of task/exam	Trial

12 c) (i) D midpoint of AB

$$\therefore D = \left(\frac{-10+4}{2}, \frac{0+6}{2} \right) = (-3, -3)$$

(ii) distance between D (-3, -3) and C (0, 4)

$$d = \sqrt{(-3-0)^2 + (-3-4)^2} = \sqrt{9+49} = \sqrt{58} \text{ units}$$

$$(iii) M_{AC} = \frac{-6-4}{4-0} = -\frac{5}{2}$$

$$y\text{-int} = 4$$

$$\therefore y = -\frac{5}{2}x + 4$$

$$2y = -5x + 8$$

$$5x + 2y - 8 = 0$$

$$(iv) M_{BC} = \frac{4-0}{0-10} = -\frac{2}{5}$$

$$M_{AC} \times M_{BC} = -\frac{5}{2} \times -\frac{2}{5} = 1$$

$\therefore$ line h $\perp$ line k

(v) centre of circle is D (-3, -3)

radius = distance of BD

$$= \sqrt{(-10-(-3))^2 + (0-(-3))^2} = \sqrt{49+9} = \sqrt{58}$$

$\therefore$ circle with centre (-3, -3) and radius $\sqrt{58}$

$$(x+3)^2 + (y+3)^2 = 58$$

(vi) radius $\sqrt{58}$

$$\text{Area} = \pi r^2 = \pi \times (\sqrt{58})^2 = 58\pi \text{ units}^2$$

Question 13

$$a) \ln\left(\frac{x^2+9}{5x}\right) = \ln 2$$

$$\frac{x^2+9}{5x} = 2$$

$$x^2+9 = 10x$$

$$x^2-10x+9 = 0$$

$$(x-1)(x-9) = 0$$

$$x = 1 \text{ or } 9$$

Academic Year	12	Calendar Year	2019
Course	2Unit	Name of task/exam	Trial

13 b) (i) $y' = 6x^2 - 18x + 12$

stat pts $y' = 0$

$$6x^2 - 18x + 12 = 0$$

$$x^2 - 3x + 2 = 0$$

$$(x-2)(x-1) = 0$$

$$x = 2 \text{ or } 1$$

$$y'' = 12x - 18$$

Nature of pts.

$$\text{at } x = 2 \quad y'' = 12 \times 2 - 18 = 6 > 0 \therefore \text{min}$$

$$\text{at } x = 1 \quad y'' = 12 \times 1 - 18 = -6 < 0 \therefore \text{max}$$

$\therefore$ min pt at (2, -1)

max pt at (1, 0)

(ii) Pt of inf. $y'' = 0$

$$12x - 18 = 0$$

$$12x = 18$$

$$x = 1.5$$

$$\therefore y = 2 \times 1.5^3 - 9 \times 1.5^2 + 12 \times 1.5 - 5 = -0.5$$

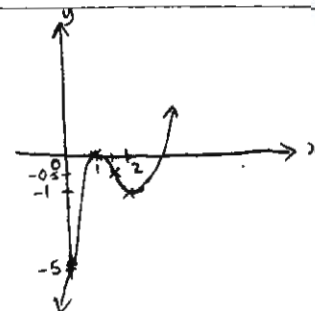
$\therefore$ possible pt of inf. at (1.5, -0.5)

Test	x	1	1.5	2
	y''	-	0	+

change in concavity

$\therefore$ pt of inflexion at (1.5, -0.5)

(iii)



$$c) \lim_{x \rightarrow 2} \frac{x^3 - x^2}{2 - x} = \lim_{x \rightarrow 2} \frac{x(2+x)(2-x)}{2-x} = \lim_{x \rightarrow 2} x(2+x) = 2(2+2) = 8$$

d) (i) $t = 0$ (2000) $P = 19$ mill.
 $t = 19$ (2019) $P = 25$ mill

$$P = Ae^{kt}$$

$$\frac{dP}{dt} = kAe^{kt}$$

$$= kP \text{ since } P = Ae^{kt}$$

(ii) Sub $P = 19$ and $t = 0$ into $P = Ae^{kt}$

$$19 = Ae^{k \times 0}$$

$$\therefore A = 19 \text{ million}$$

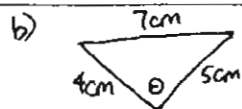
Academic Year	12	Calendar Year	2019
Course	2 Unit	Name of task/exam	Trial

13d) (iii) $P = 19e^{19k}$
 sub $t=19$, $P=25$
 $e^{19k} = \frac{25}{19}$
 $\ln e^{19k} = \ln\left(\frac{25}{19}\right)$
 $19k = \ln\left(\frac{25}{19}\right)$
 $k = \frac{\ln\left(\frac{25}{19}\right)}{19}$
 $= 0.01444\dots$
 $\doteq 0.0144$

(iv) In 2050 $t=50$
 sub $t=50$ into P
 $P = 19e^{50 \times 0.0144}$
 $\doteq 39 \text{ million} < 50 \text{ million}$
 $\therefore$ incorrect

Question 14:

a) $f'(x) = -2e^{1-2x}$
 at $x = \frac{1}{2}$, $f'\left(\frac{1}{2}\right) = -2e^{1-2 \times \frac{1}{2}}$
 $= -2e^0$
 $= -2$
 $\therefore$ eqn of tangent
 $y - 1 = -2\left(x - \frac{1}{2}\right)$
 $y = -2x + 2$



b) $\cos \theta = \frac{4^2 + 5^2 - 7^2}{2 \times 4 \times 5}$
 $= -\frac{1}{5}$
 $\theta = \cos^{-1}\left(-\frac{1}{5}\right)$
 $= 1.77 \text{ radians (to 2dp)}$

c) (i) Point A $-2x = -(x-3)$
 $-2x = -x+3$
 $-3 = x$

Point B $2x = -(x-3)$
 $2x = -x+3$
 $3x = 3$
 $x = 1$

(ii) $x \leq -3$, $x \geq 1$

d) (i) stop flowing $R=0$
 $3\pi t(8-t) = 0$
 $t=0$ or $t=8$
 $\therefore$ after 8 seconds.

Academic Year	12	Calendar Year	2019
Course	2 Unit	Name of task/exam	Trial

14d) (ii) empty when $V=0$
 where $V =$ volume of water
 in fountain at time t

$V = \int R$
 $= \int 24\pi t - 3\pi t^2 dt$
 $= \frac{24\pi t^2}{2} - \frac{3\pi t^3}{3} + C$
 $= 12\pi t^2 - \pi t^3 + C$

at $t=0$, $V=0$

$\therefore V=0$

$V = 12\pi t^2 - \pi t^3$

Empty when $V=0$

$12\pi t^2 - \pi t^3 = 0$

$\pi t^2(12-t) = 0$

$t=0$ or 12

$\therefore$ empty after 12 seconds

(iii) Sub $t=12$ into R

$R = 3\pi \times 12(8-12)$

$= -144\pi \text{ L/s}$

e) $\frac{dx}{dt} = 1 + \frac{3}{3t+1}$

At rest $\frac{dx}{dt} = 0$

$1 + \frac{3}{3t+1} = 0$

$\frac{3}{3t+1} = -1$

$3 = -3t-1$

$3t = -4$

$t = -\frac{4}{3} < 0$

$\therefore$ never at rest.

(ii) At $t=2$, $x = 2 + \ln 7$

At $t=3$, $x = 3 + \ln 10$

dist travelled

$= (3 + \ln 10) - (2 + \ln 7)$

$= 1.4 \text{ cm (to 1 decimal place)}$

(iii) $\frac{d^2x}{dt^2} = -3(3t+1)^{-2} \cdot 3$
 $= -\frac{9}{(3t+1)^2}$

Academic Year	12	Calendar Year	2019
Course	2 Unit	Name of task/exam	Trial

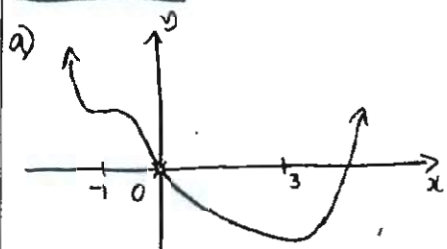
14e (iv) For $t > 0$:

$$a = -\frac{9}{(3t+1)^2} < 0 \text{ for all } t$$

since $(3t+1)^2$ always positive, $\therefore$ acceleration is always negative.

$v = 1 + \frac{3}{3t+1}$ which is always greater than 0. As a and v are opposite signs, the particle is slowing down.

Question 15



Horizontal Pt of Inflection at $x = -1$

Min pt at $x = 3$

$$\begin{aligned} \text{b) } 2(1 - \sin^2 x) + \sin x &= 2 \\ 2 - 2\sin^2 x + \sin x - 2 &= 0 \\ \sin x(1 - 2\sin x) &= 0 \end{aligned}$$

$$\begin{aligned} \sin x &= 0 \text{ or } \sin x = \frac{1}{2} \\ x &= 0, \pi, \text{ or } x = \frac{\pi}{6}, \frac{5\pi}{6} \end{aligned}$$

$$\text{c) (i) } y = e^{ax} - 2$$

$$\text{Sub } x = \pi \quad y = 0$$

$$0 = e^{a\pi} - 2$$

$$e^{a\pi} = 2$$

$$\ln e^{a\pi} = \ln 2$$

$$a\pi = \ln 2$$

$$a = \frac{\ln 2}{\pi}$$

$$\text{(ii) } A = \int_0^\pi \sin x - (e^{ax} - 2) dx$$

$$= \int_0^\pi \sin x - e^{\frac{\ln 2}{\pi} x} + 2 dx$$

$$= \left[-\cos x - \frac{\pi e^{\frac{\ln 2}{\pi} x}}{\ln 2} + 2x \right]_0^\pi$$

$$\begin{aligned} &= (-\cos \pi - \frac{\pi e^{\ln 2}}{\ln 2} + 2\pi) \\ &\quad - (-\cos 0 - \frac{\pi e^0}{\ln 2} + 0) \end{aligned}$$

$$= -(-1) - \frac{2\pi}{\ln 2} + 2\pi + 1 + \frac{\pi}{\ln 2}$$

$$= 2 + 2\pi - \frac{\pi}{\ln 2}$$

$$= 3.75 \text{ (to 3 sig fig)}$$

Academic Year	12	Calendar Year	2019
Course	2 Unit	Name of task/exam	Trial

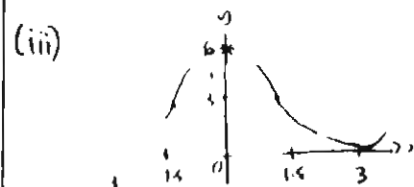
$$15 \text{ d) (i) } \frac{3}{6} (2 + 4 \times 3.1 + 4)$$

$$+ \frac{3}{6} (2 + 4 \times 3.1 + 4)$$

$$= 18.4 \text{ m}^2$$

$$\text{(ii) } \frac{3}{4} (2 + 3.1) \times 2 + 2 \times \frac{3}{4} (4 + 3.1)$$

$$= 18.3 \text{ m}^2$$



$$\int_0^3 3 \cos x dx = [3x + 3 \sin x]_0^3$$

$$(9 + 3 \sin 3) - (-9 + 3 \sin(-3))$$

$$= 18 + 3 \sin 3 - 3 \sin(-3)$$

$$= 18.85$$



When you draw a parabola through the points and straight lines between each point the gap between them is quite small.

Question 16

a) (i) x -coord of P is 4

Sub $x = 4$ into y

$$y = 1 + 2\sqrt{4}$$

$$= 5$$

$\therefore B(0, 5)$

$$\text{(ii) } V = \pi \int x^2 dy$$

$$y = 1 + 2\sqrt{x}$$

$$y - 1 = 2\sqrt{x}$$

$$\frac{y-1}{2} = \sqrt{x}$$

$$\frac{(y-1)^2}{4} = x$$

$$\frac{(y-1)^4}{16} = x^2$$

$$V = \pi \int_1^5 \frac{(y-1)^4}{16} dy$$

$$= \frac{\pi}{16} \int_1^5 (y-1)^4 dy$$

$$= \frac{\pi}{16} \left[\frac{(y-1)^5}{5} \right]_1^5$$

$$= \frac{\pi}{16} \left(\frac{4^5}{5} - 0 \right)$$

$$= \frac{64\pi}{5} \text{ or } 12.8\pi$$

Academic Year	12	Calendar Year	2019
Course	2 Unit	Name of task/exam	Trial

$$16) b) \int \sin^2 x \cos x = \frac{1}{3} \sin^3 x + C$$

$$\text{sub } x = \frac{\pi}{2} \quad y = 1$$

$$\frac{1}{3} \sin^3 \frac{\pi}{2} + C = 1$$

$$\frac{1}{3} + C = 1$$

$$C = \frac{2}{3}$$

$$\therefore f(x) = \frac{1}{3} \sin^3 x + \frac{2}{3}$$

$$c) (i) \text{ Vol. of cone} = \frac{1}{3} \pi r^2 h$$

$$r^2 = 12^2 - h^2$$

$$\therefore V = \frac{1}{3} \pi (12^2 - h^2) h$$

$$= \frac{1}{3} \pi h (144 - h^2)$$

$$V = 48\pi h - \frac{\pi h^3}{3}$$

$$(ii) \frac{dV}{dh} = 48\pi - \pi h^2$$

$$\text{stat pts } 48\pi - \pi h^2 = 0$$

$$h^2 = 48$$

$$h = \pm \sqrt{48}$$

$$h > 0 \therefore h = \sqrt{48}$$

$$\frac{d^2V}{dh^2} = -2\pi h$$

$$\text{when } h = \sqrt{48}, \frac{d^2V}{dh^2} = -2\sqrt{48}\pi < 0$$

$$\therefore \text{max when } h = \sqrt{48}$$

$\therefore$ Max volume

$$= 48\pi \times \sqrt{48} - \frac{\pi \times (\sqrt{48})^3}{3}$$

$$= 696.5 \text{ cm}^3 \text{ (to 1 dp)}$$

$$d) \int_0^3 f''(x) dx = -10$$

$$f(3) - f(0) = -10$$

$$\text{Given } f(0) = 12$$

$$\therefore f(3) - 12 = -10$$

$$f(3) = 2$$

Similarly,

$$f(6) - f(3) = -3.65$$

$$f(6) - 2 = -3.65$$

$$f(6) = -1.65$$

And

$$f(7) - f(6) = 3.65$$

$$f(7) - (-1.65) = 3.65$$

$$f(7) + 1.65 = 3.65$$

$$f(7) = 2$$

Academic Year	12	Calendar Year	2019
Course	2 Unit	Name of task/exam	Trial

$$16d) (ii) g(x) = (f(x))^2$$

$$\therefore g(7) = (f(7))^2$$

$$= (2)^2$$

$$= 4$$

$\therefore (7, 4)$ on the line

$$g'(x) = 2f'(x)f(x)$$

$$\therefore g'(7) = 2f'(7)f(7)$$

$$= 2 \times 14 \times 2$$

$$= 56$$

$\therefore$ equation of tangent with m 56 and passes through (7, 4):

$$y - 4 = 56(x - 7)$$

$$y = 56x - 392 + 4$$

$$y = 56x - 388$$